Nathan Wallheimer

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Education

2022–present	PhD in Computer Science (in progress), Weizmann Institute of Science. Advisor: Prof. Amir Abboud. Thesis: *Structure versus Pseudorandomness in Fine-Grained Complexity.*
2018–2021	MSc in Computer Science, University of Haifa. Advisor: Prof. Oren Weimann.
2014–2018	BSc in Computer Science, University of Haifa.

Professional Experience and Activities

July 2024	Teaching Assistant, DIMACS, Rutgers University.
2017–2021	Teaching Assistant, Department of Computer Science, University of Haifa.
2016–2018	Private tutor, academic support program for BSc students experiencing academic difficulties, University of Haifa.

Conference presentations: ISAAC 2022, ITCS 2023, HALG 2023, ESA 2024, SODA 2025, ITCS 2026, and HALG 2026.

Reviewer for:

- IEEE Symposium on Foundations of Computer Science (FOCS).
- ACM-SIAM Symposium on Discrete Algorithms (SODA).
- International Colloquium on Automata, Languages, and Programming (ICALP).
- Innovations in Theoretical Computer Science Conference (ITCS).
- European Symposium on Algorithms (ESA).
- Scandinavian Symposium on Algorithm Theory (SWAT).
- International Symposium on Mathematical Foundations of Computer Science (MFCS).
- Annual Symposium on Combinatorial Pattern Matching (CPM).

Teaching and Academic Service

2022–present	Algorithms Seminar Organizer, Weizmann Institute of Science. Select speakers, coordinate logistics, and host guests.
July 2024	DIMACS Tutorial on Fine-Grained Complexity, Rutgers University.
2018–2021	Data Structures, University of Haifa.
2017–2021	Advanced Data Structures and Probability, University of Haifa.

Honours and Awards

2021	Dean's List, University of Haifa (MSc in Computer Science).
2019–2020	Excellence scholarship, University of Haifa.

Languages: Hebrew, English, French.

Publications

1. Amir Abboud, Shyan Akmal, Ron Safier, and Nathan Wallheimer.
   Lower Bounds for Structured OMv, APSP, and Triangle Detection.
   Innovations in Theoretical Computer Science Conference (ITCS 2027). To appear.

2. Amir Abboud, Ron Safier, and Nathan Wallheimer.
   Equivalent Dichotomies for Triangle Detection in Subgraph, Induced, and Colored H-Free Graphs.
   In *34th Annual European Symposium on Algorithms (ESA 2026)*, LIPIcs 388, pp. 64:1–64:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ESA.2026.64.

3. Keren Censor-Hillel, Tomer Even, Virginia Vassilevska Williams, and Nathan Wallheimer.
   Witness-Sensitive Detection of Induced Diamonds.
   In *53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)*, LIPIcs 374, pp. 52:1–52:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ICALP.2026.52.

4. Amir Abboud, Ron Safier, and Nathan Wallheimer.
   Triangle Detection in H-Free Graphs.
   In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, LIPIcs 362, pp. 1:1–1:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi:10.4230/LIPIcs.ITCS.2026.1.

5. Amir Abboud, Nick Fischer, Ron Safier, and Nathan Wallheimer.
   Recognizing Sumsets is NP-Complete.
   In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 4484–4506. SIAM, 2025. doi:10.1137/1.9781611978322.153.

6. Amir Abboud and Nathan Wallheimer.
   Worst-Case to Expander-Case Reductions: Derandomized and Generalized.
   In *32nd Annual European Symposium on Algorithms (ESA 2024)*, LIPIcs 308, pp. 4:1–4:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024. doi:10.4230/LIPIcs.ESA.2024.4.

7. Amir Abboud and Nathan Wallheimer.
   Worst-Case to Expander-Case Reductions.
   In *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*, LIPIcs 251, pp. 1:1–1:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023. doi:10.4230/LIPIcs.ITCS.2023.1.

8. Shay Mozes, Nathan Wallheimer, and Oren Weimann.
   Improved Compression of the Okamura-Seymour Metric.
   In *33rd International Symposium on Algorithms and Computation (ISAAC 2022)*, LIPIcs 248, pp. 27:1–27:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.ISAAC.2022.27.